

Weekly Update

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Last Update: September 20, 2026

1 Main Findings

1.1 DB-Sorter

quick background. DB-sorter uses the sorting characteristic of DB flow to find ground state or excited states of a hamiltonian H using a diagonal matrix D that has an engineered order in its diagonal elements (usually increasing). **We can derive Grover's algorithm using DB-Sorter**

Findings:

- Grover's algorithm can be improved drastically by replacing its normal reflection method with other diagonal D 's. The traditional $I - 2|0\rangle\langle 0|$ Operator needs multiple control phase shift which results in $\mathcal{O}(n^2)$ CNOT's this can be avoided. Also overshoot issue can be managed this way. See [GitHub](#) for code and [3.1.1](#) for figures.

1.2 Fidelity Witness

quick background. I am working on finding a new fidelity witness. This week effort was through the intermediate step:

$$\forall O : F(P, Q) \geq F(P, O)^2 + F(O, Q)^2 - 1 \quad (1)$$

It means, for creating a witness, we need to find an O for which we can easily compute any $F(O, Q)$.

Findings.

- I assessed Gaussian O 's, but with what we already have, they do not seem promising candidates. See [3.2.1](#)
- Using **Classical Shadows** we can have a concise classical representation of P and Q . They also give us a reasonable approximation for $F(O, P)$ classically. This reduces finding O to a classical optimization. [3.2.2](#)
- This bound is pretty loose. Intuitively, we need to find an O close enough to both P and Q (e.g $F(P, O) > 0.7$) otherwise the right hand side of the inequality is negative which is trivial.

2 Plan For Next Week(s)

2.1 DB-sorter

- finding proper D that guarantees monotone error drop in $\frac{d}{dt}(\langle k| H(t) |k\rangle - E_k)^2$ for all k 's and not just ground state
- If U is the unitary transforming $H(0)$ to $H(t)$, finding the fidelity $U|0\rangle$ and the ground state(s) to see if the prepared state constantly converges to a single ground state or is a superposition of degenerate states. Same for the k 'th excited state
- How many steps for Grover's algorithm?

2.2 Fidelity Witness

- explore classical shadows even more and see how far we can go with them

3 Details

3.1 DB-sorter

3.1.1 Grover's algorithm and DB-sorter

In last week's update we see that we can derive Grover's algorithm using DB-sorter. Here I implemented this comparison. This change allow us to change D and s and come up with variations of Grover's algorithm. Here we compare traditional Grover's algorithm with the case that we replace $I - 2|0\rangle\langle 0|$ projection with a diagonal matrix with ascending elements and $s = 1$ with $s = .5$. There are two benefits and one draw-back:

1. The depth of the circuit drops significantly. It especially shows up as the number of qubit grows and qubit connections drop as we replace CNOT gates with single-qubit gates.
2. We easen up on the overshoot problem
3. The number of steps to convergence is harder to calculate?

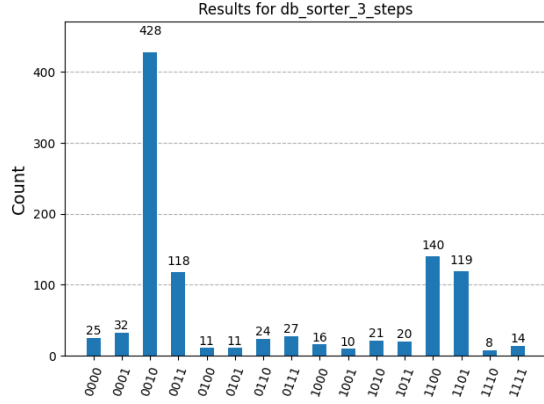
In [Figure 1](#) you see how overshoot happens with reflection variation (top row) but not with the monotone D (bottom row)

In [Figure 2](#) you can compare the grover's circuit size in traditional and our variation for the same number of iterations

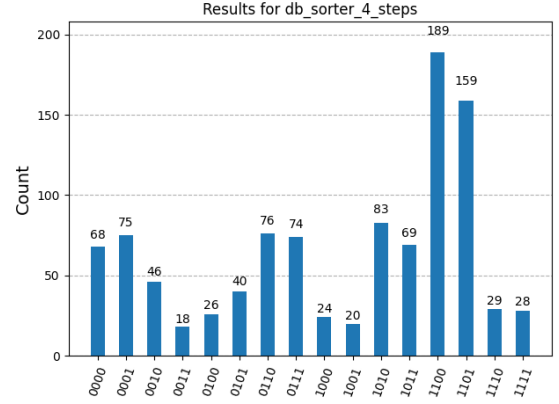
3.2 Fidelity Witness

3.2.1 Gaussian Fidelity Variational

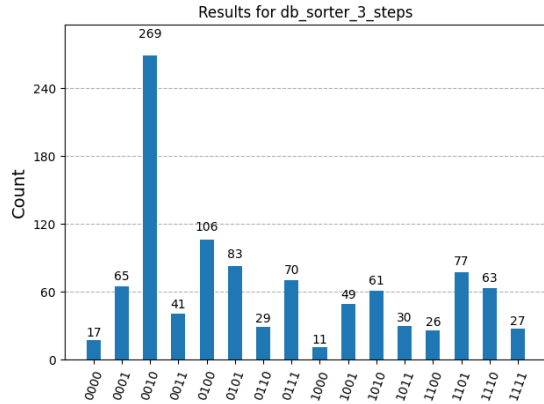
We already have a Gaussian witness. In [eq-1](#), therefore, it is tempting to have a variational optimization in the Gaussian state space. However, it will run into an issue as our witness relies on $\mathcal{W}_\omega = (\mathbf{1} - n_\omega)$. We know that for states with a Hamming distance more than one with ω , the lower bound can be negative, therefore the square would change the direction of the inequality and does not fit our use case.



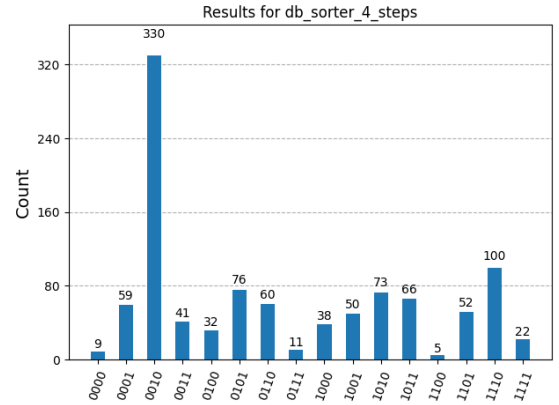
(a) Traditional Grover converged.



(b) Traditional Grover lost convergence.

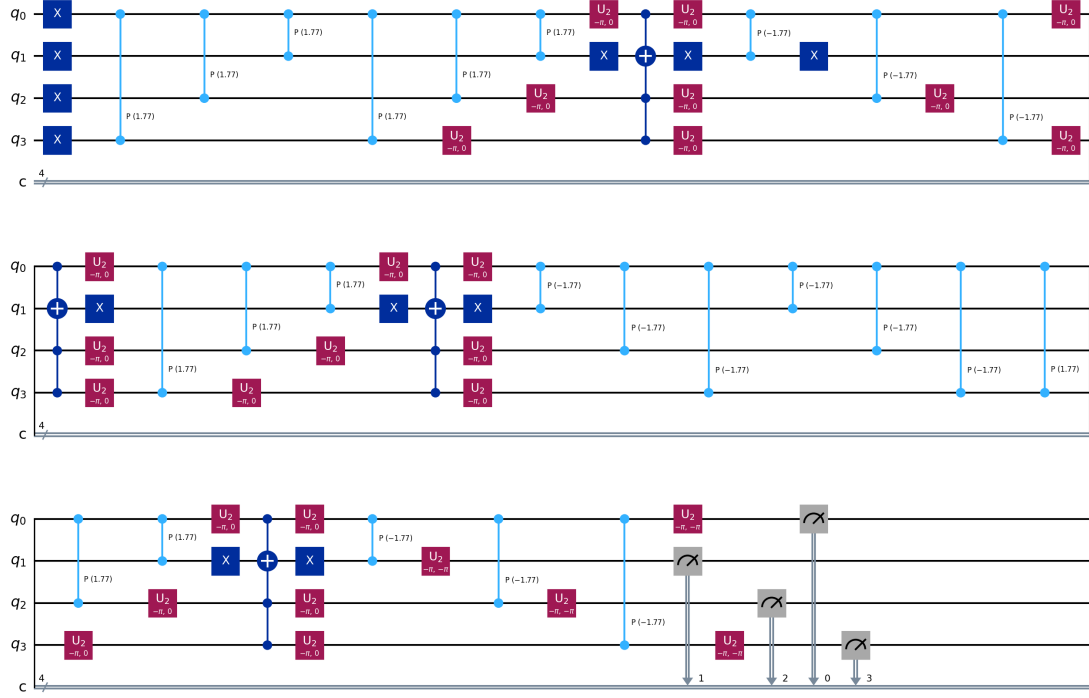


(c) Modified Grover converged.



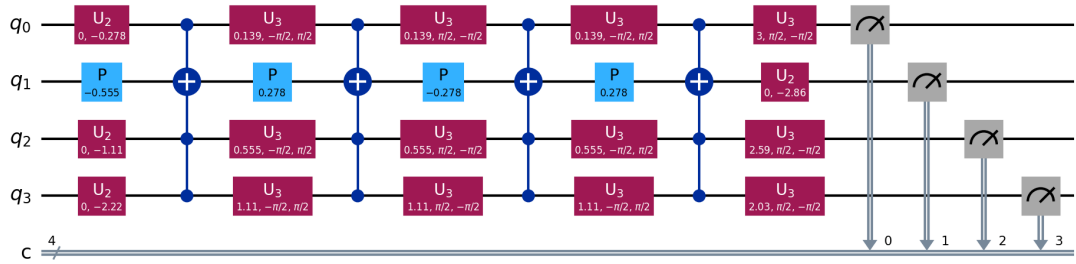
(d) Modified Grover is more resistance to divergence.

Figure 1: Convergence and divergence in Grover's variations. Marked state: $|0010\rangle$



(a) Larger Circuit size due to CNOTs required for $I - 2|0\rangle\langle 0|$.

Global Phase: 3.3757175204217313



(b) Shallower circuit due to replacing CNOTs with single-qubit gates

Figure 2: We can control Grover's depth by controlling D in DB-Sorter

3.2.2 Classical Shadow Optimization

Classical shadows are shown to be useful in capturing major information of state ρ in a set of classical snapshots and use them to estimate $Tr(O\rho)$ for any observable O . However the sample size for this relies on how complex O is. Generally, we need exponentially many samples from ρ . However for specific classes of observables (e.g separable, unentangled O) we can do so polynomially. The advantage of this method is that we do not need to know O in advance; we can store many classical snapshots of ρ and use them for the observables that come to our way. This way, as a witness for $F(P, Q)$, we can capture many snapshots from P and Q and classically optimize the left hand side of the equation $\hat{F}(P, O)^2 + \hat{F}(O, Q)^2 - 1$ where $\hat{F}$ is an estimate of fidelity using classical shadow. The optimization is a classical gradient descend (with normalization in each step to make sure O stays a valid density matrix). The problem here is finding a proper O that is close enough to both P and Q otherwise the sum of squares is less than 1 and only thing we get will be $F(P, Q) > 0$. **Similar work:** The author of classical shadow method have themselves [published a paper](#) for fidelity estimation between any states P (prepared) and Q (target) as long as at least one of them is pure. My method has two advantage and one disadvantage compared to theirs:

1. pro: my method can handle both P and Q to be mixed states
2. pro: their method requires knowledge to all coefficients of the target aka $\langle k|Q|k\rangle$ which *usually* requires either exponential copies of Q or exponential memory to store these values
3. con: my method assumes we can produce polynomial copies from the target state while theirs rely only on 1 (i mean they have the coefficients, so...)

The code is attached as a zip file but because for now I am still dealing with finding a better O I still haven't pushed it to the repo as a code structure.